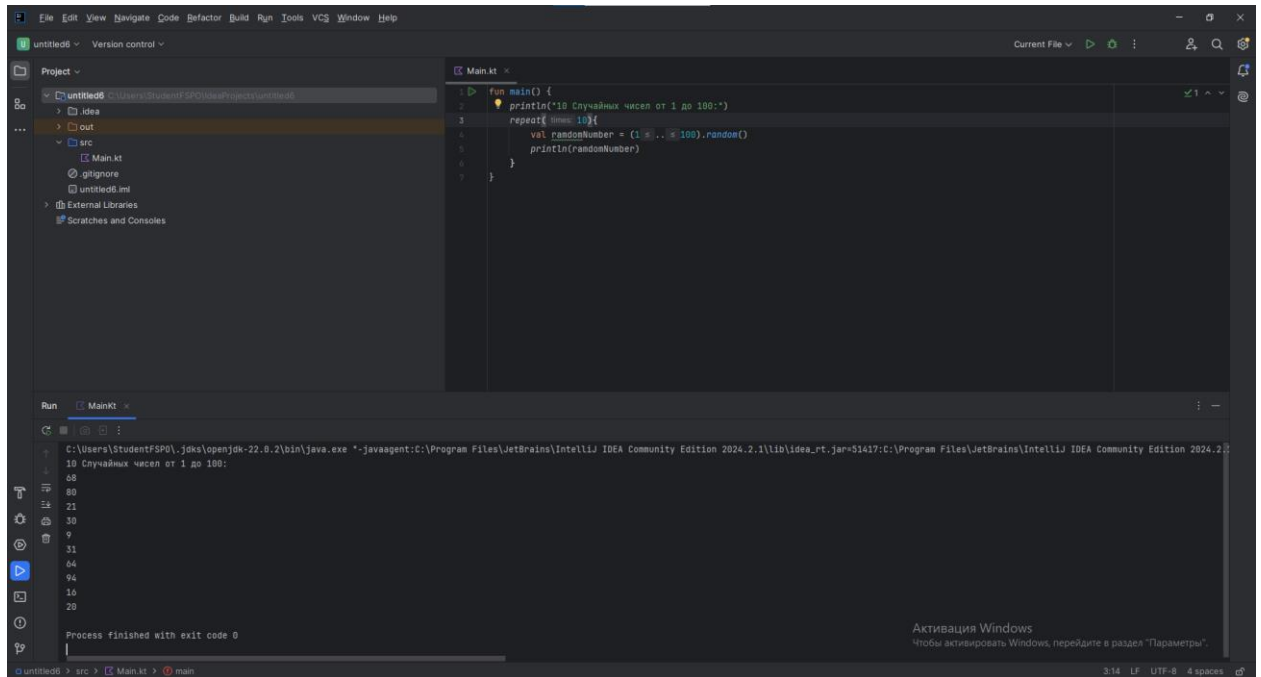


1)



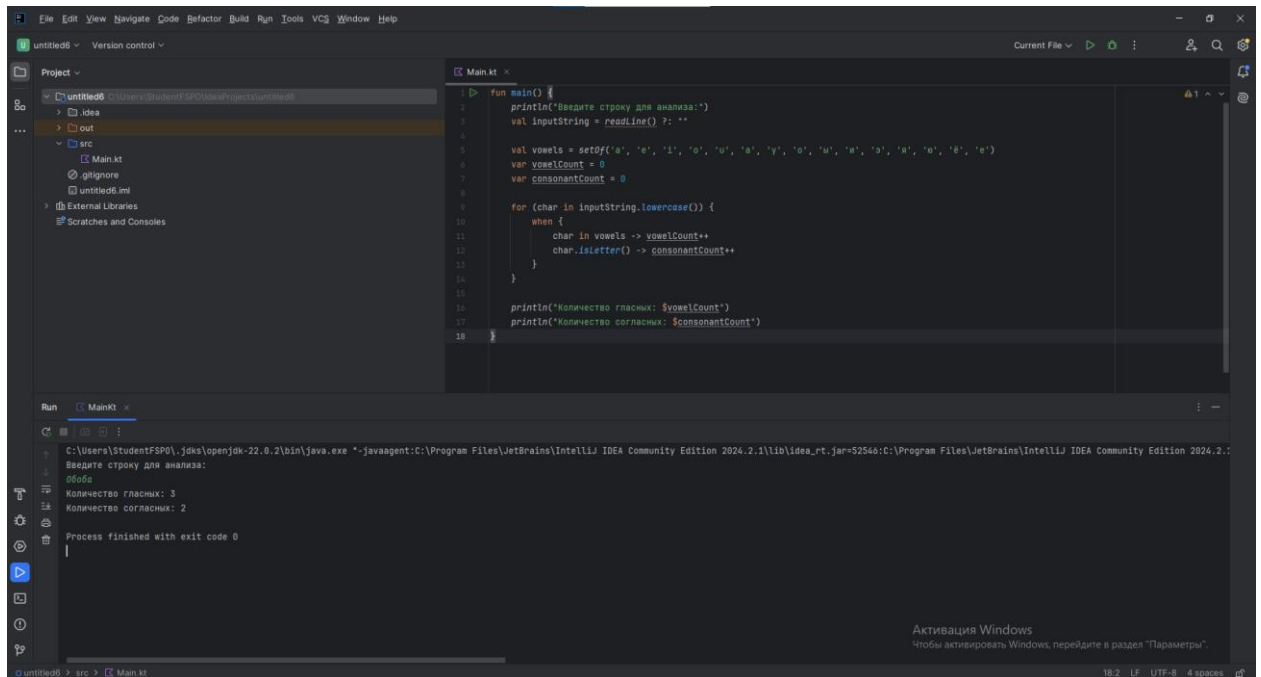
The screenshot shows the IntelliJ IDEA interface. The Project view on the left shows a project named 'untitled6' with a source file 'Main.kt'. The editor displays the following Kotlin code:

```
1 fun main() {  
2     println("10 Случайных чисел от 1 до 100:")  
3     repeat(10){  
4         val randomNumber = (1..100).random()  
5         println(randomNumber)  
6     }  
7 }
```

The Run window at the bottom shows the output of the program:

```
C:\Users\StudentSP0\j\jdk\openjdk-22.0.2\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1\lib\idea_rt.jar=51417:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1  
10 Случайных чисел от 1 до 100:  
68  
21  
30  
9  
31  
64  
94  
16  
20  
Process finished with exit code 0
```

2)



The screenshot shows the IntelliJ IDEA interface. The Project view on the left shows a project named 'untitled6' with a source file 'Main.kt'. The editor displays the following Kotlin code:

```
1 fun main() {  
2     println("Введите строку для анализа:")  
3     val inputString = readLine() ?: ""  
4  
5     val vowels = setOf('a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U', 'a', 'e', 'i', 'o', 'u')  
6     var vowelCount = 0  
7     var consonantCount = 0  
8  
9     for (char in inputString.lowercase()) {  
10        when {  
11            char in vowels -> vowelCount++  
12            char.isLetter() -> consonantCount++  
13        }  
14    }  
15  
16    println("Количество гласных: $vowelCount")  
17    println("Количество согласных: $consonantCount")  
18 }
```

The Run window at the bottom shows the output of the program:

```
C:\Users\StudentSP0\j\jdk\openjdk-22.0.2\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1\lib\idea_rt.jar=52546:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1  
Введите строку для анализа:  
abca  
Количество гласных: 3  
Количество согласных: 2  
Process finished with exit code 0
```

3)

```
fun main() {  
    val exchangeRates = mapOf(  
        "USD" to mapOf("EUR" to 0.85, "RUB" to 75.50),  
        "EUR" to mapOf("USD" to 1.18, "RUB" to 89.25),  
        "RUB" to mapOf("USD" to 0.013, "EUR" to 0.011)  
    )  
  
    println("Доступные валюты: USD, EUR, RUB")  
    println("Введите исходную валюту (например, USD):")  
    val fromCurrency = readLine()?.uppercase() ?: ""  
  
    println("Введите целевую валюту (например, EUR):")
```

```

val toCurrency = readLine()?.uppercase() ?: ""

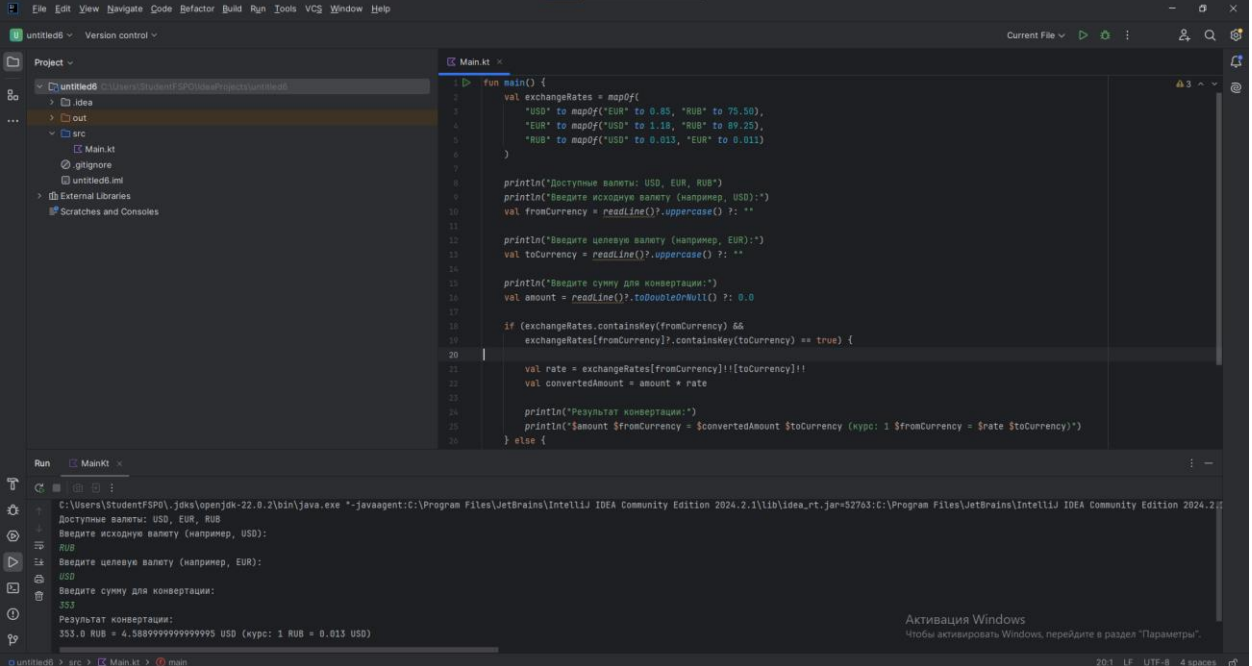
println("Введите сумму для конвертации:")
val amount = readLine()?.toDoubleOrNull() ?: 0.0

if (exchangeRates.containsKey(fromCurrency) &&
    exchangeRates[fromCurrency]?.containsKey(toCurrency) == true) {

    val rate = exchangeRates[fromCurrency]!![toCurrency]!!
    val convertedAmount = amount * rate

    println("Результат конвертации:")
    println("$amount $fromCurrency = $convertedAmount $toCurrency (курс:
1 $fromCurrency = $rate $toCurrency)")
} else {
    println("Ошибка: Неподдерживаемая валюта или отсутствует курс
конвертации")
}
}

```



The screenshot shows the IntelliJ IDEA IDE with a Kotlin project. The code in the editor implements a currency conversion program. It defines a map of exchange rates and reads user input for the source currency, target currency, and amount. It then calculates the converted amount and prints the result. The Run window shows the execution output, including the prompt for the source currency (USD), the prompt for the target currency (EUR), and the final conversion result: 553.0 RUB = 4.5889999999999995 USD (курс: 1 RUB = 0.013 USD).

4)

```

fun main() {
    println("Введите первую строку:")
    val str1 = readLine() ?: ""

    println("Введите вторую строку:")
    val str2 = readLine() ?: ""

    if (areAnagrams(str1, str2)) {
        println("Строки являются анаграммами")
    } else {
        println("Строки НЕ являются анаграммами")
    }
}

fun areAnagrams(str1: String, str2: String): Boolean {
    val s1 = str1.replace("\\s".toRegex(), "").lowercase()
    val s2 = str2.replace("\\s".toRegex(), "").lowercase()

    if (s1.length != s2.length) return false
}

```

```
val charCount1 = mutableMapOf<Char, Int>()
val charCount2 = mutableMapOf<Char, Int>()

for (char in s1) {
    charCount1[char] = charCount1.getOrDefault(char, 0) + 1
}

for (char in s2) {
    charCount2[char] = charCount2.getOrDefault(char, 0) + 1
}

return charCount1 == charCount2
}

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
untitled6 Version control
Project:
  untitled6 C:\Users\Student\SP\Idea\projects\untitled6
  > idea
  > out
  > src
    Main.kt
    .gitignore
    untitled6.iml
  > External Libraries
  > Scratches and Consoles
Main.kt
1 fun main() {
2     println("Введите первую строку:")
3     val str1 = readLine() ?: ""
4
5     println("Введите вторую строку:")
6     val str2 = readLine() ?: ""
7
8     if (areAnagrams(str1, str2)) {
9         println("Строки являются анаграммами")
10    } else {
11        println("Строки НЕ являются анаграммами")
12    }
13 }
14
15 fun areAnagrams(str1: String, str2: String): Boolean {
16     val s1 = str1.replace("\\s".toRegex(), "").lowercase()
17     val s2 = str2.replace("\\s".toRegex(), "").lowercase()
18
19     if (s1.length != s2.length) return false
20
21     val charCount1 = mutableMapOf<Char, Int>()
22     val charCount2 = mutableMapOf<Char, Int>()
23
24     for (char in s1) {
25         charCount1[char] = charCount1.getOrDefault(char, 0) + 1
26     }
27
28     for (char in s2) {
29         charCount2[char] = charCount2.getOrDefault(char, 0) + 1
30     }
31 }
Run Main.kt
C:\Users\Student\SP\Idea\openjdk-22.0.2\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1\lib\idea_rt.jar=52997:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1\bin -Dfile.encoding=UTF-8
Введите первую строку:
Арсений
Введите вторую строку:
Александр
Строки НЕ являются анаграммами
Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".
16:59 LF UTF-8 4 spaces
```

```
5) fun main() {
    println("Введите число N для поиска простых чисел до N:")
    val n = readLine()?.toIntOrNull() ?: 0

    if (n < 2) {
        println("Нет простых чисел до $n")
        return
    }

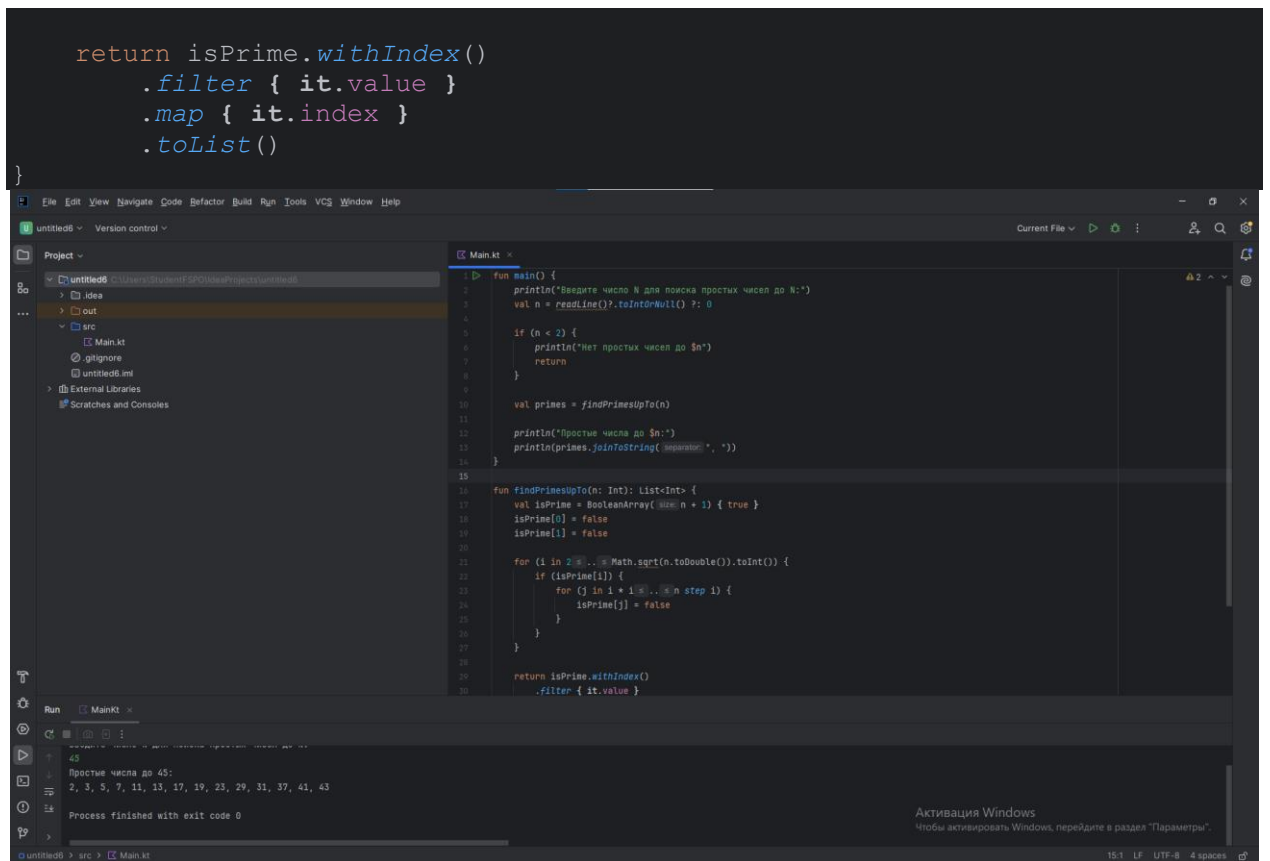
    val primes = findPrimesUpTo(n)

    println("Простые числа до $n:")
    println(primes.joinToString(", "))
}

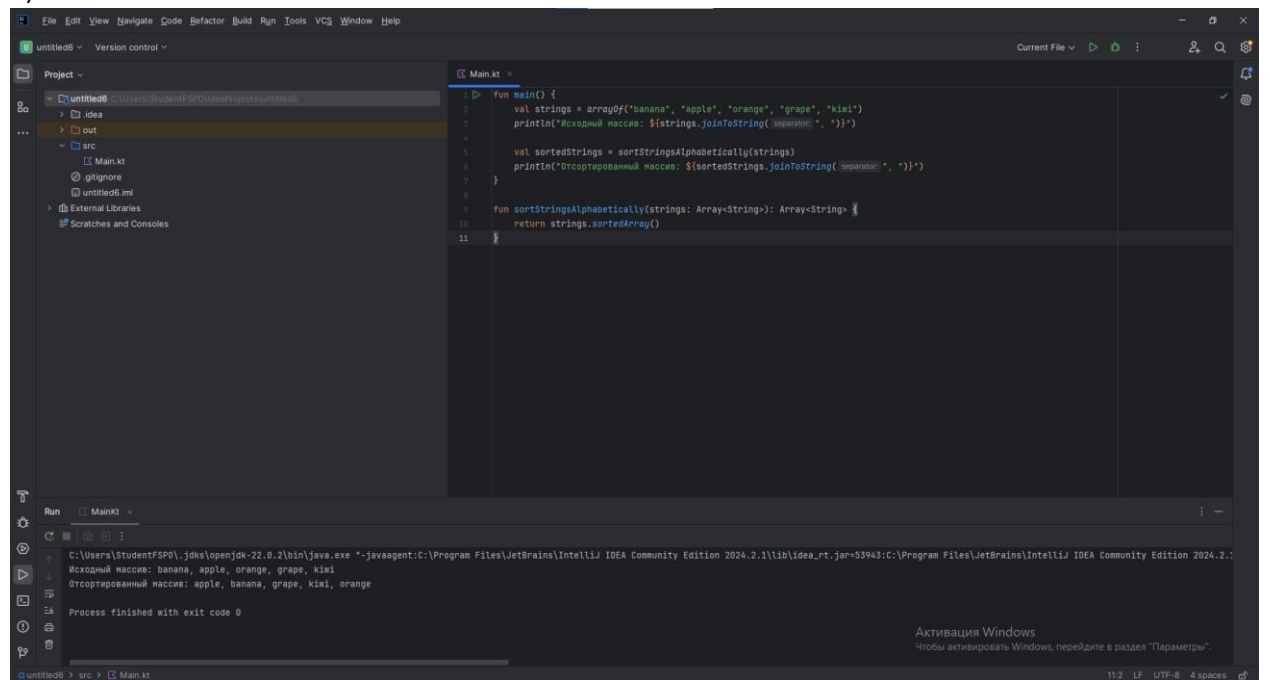
fun findPrimesUpTo(n: Int): List<Int> {
    val isPrime = BooleanArray(n + 1) { true }
    isPrime[0] = false
    isPrime[1] = false

    for (i in 2..Math.sqrt(n.toDouble()).toInt()) {
        if (isPrime[i]) {
            for (j in i * i..n step i) {
                isPrime[j] = false
            }
        }
    }

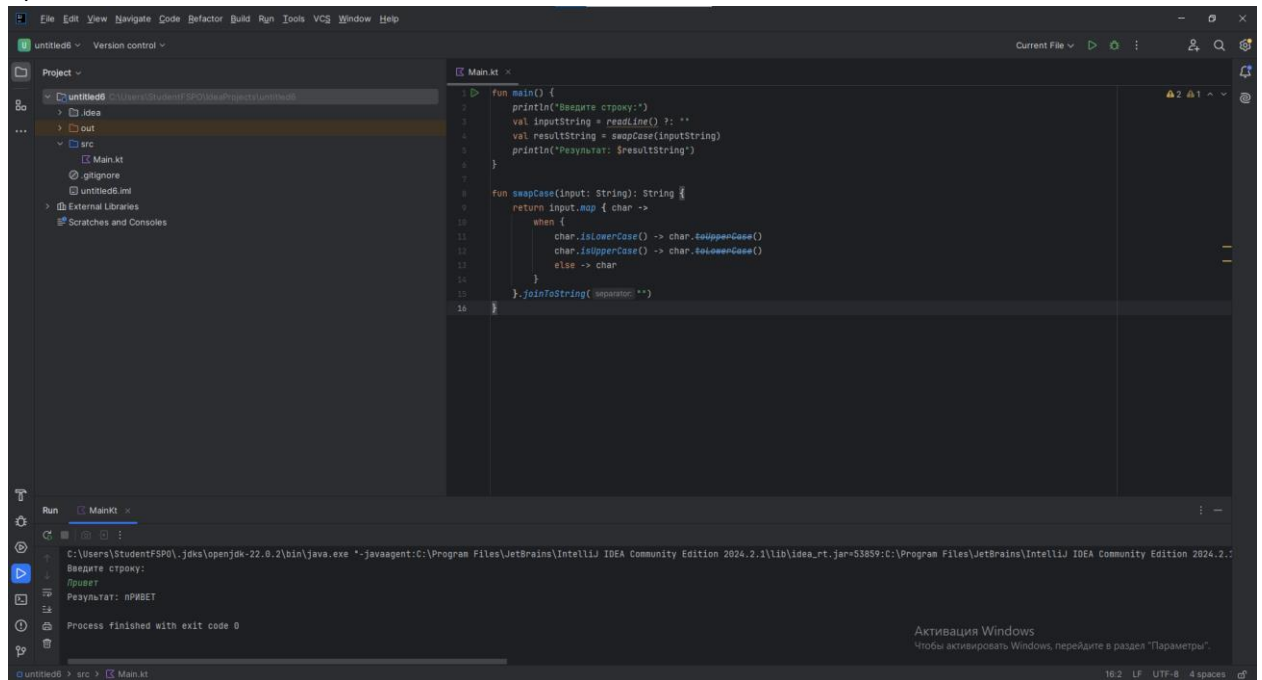
    return isPrime.filter { it }.indices.map { it }
}
```



6)



7)



```

8) import kotlin.random.Random

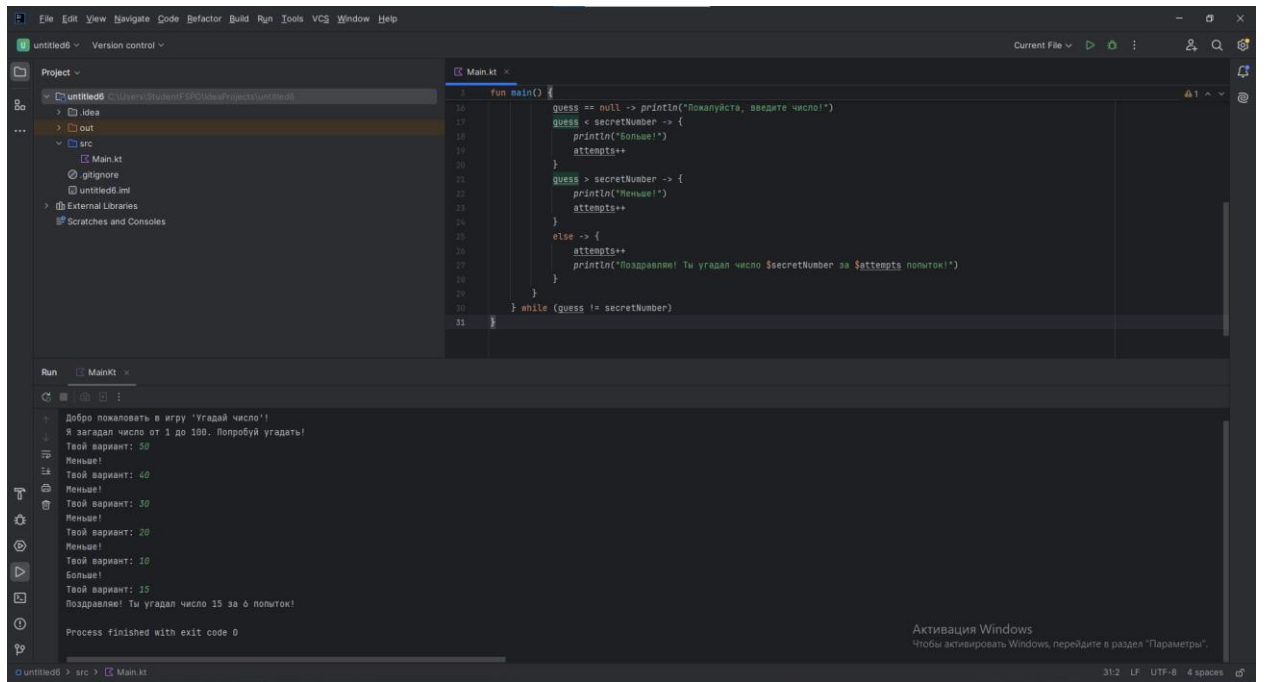
fun main() {
    val secretNumber = Random.nextInt(1, 101) // Загадываем число от 1 до 100
    var attempts = 0
    var guess: Int?

    println("Добро пожаловать в игру 'Угадай число!'")
    println("Я загадал число от 1 до 100. Попробуй угадать!")

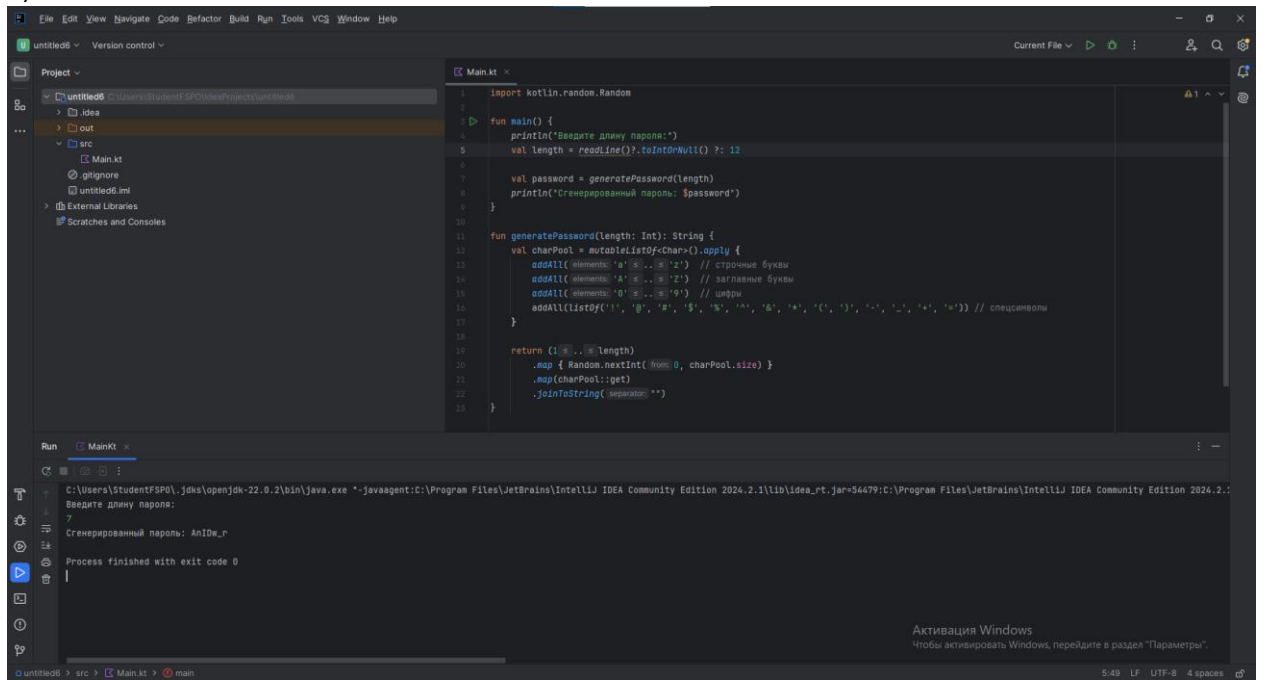
    do {
        print("Твой вариант: ")
        guess = readLine()?.toIntOrNull() // Читаем ввод пользователя

        when {
            guess == null -> println("Пожалуйста, введите число!")
            guess < secretNumber -> {
                println("Больше!")
                attempts++
            }
            guess > secretNumber -> {
                println("Меньше!")
                attempts++
            }
            else -> {
                attempts++
                println("Поздравляю! Ты угадал число $secretNumber за $attempts попыток!")
            }
        }
    } while (guess != secretNumber)
}

```



9)



10)

